



Sebastián Nájera Valencia

Researcher in Gravitation and Cosmology

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Summary

Researcher in gravitation and relativistic cosmology working on General Relativity, gravitational entropy, inhomogeneous spacetimes, modified gravity, exact solutions, and primordial black-hole models.

Education

PhD Instituto de Ciencias Nucleares, Universidad Nacional Autónoma de México (UNAM), Physics Mexico City, Mexico 2024
 Thesis: Gravitation: Cosmological Applications Of Exact Solutions Of Einstein's Equations With Inhomogeneous And Non-Trivial Mass-Energy Distributions
 Honorable Mention • Committee: Roberto A. Sussman (advisor), Celia Escamilla-Rivera, and Juan Carlos Hidalgo

MSc Instituto de Ciencias Nucleares, Universidad Nacional Autónoma de México (UNAM), Physics Mexico City, Mexico 2018
 Thesis: Geometric and physical properties of closed ever expanding dust models
 • Committee: Roberto A. Sussman (advisor), Hernando Quevedo, and Jerónimo Cortés

BSc in Physics Instituto de Ciencias, Universidad Nacional Autónoma de México (UNAM), Physics Mexico City, Mexico 2016

Experience

Instituto de Ciencias Nucleares, Universidad Nacional Autónoma de México (UNAM), Mexico City, Mexico Present
 Researcher in Gravitation and Cosmology
 Research on General Relativity, relativistic cosmology, gravitational entropy, exact solutions of Einstein's equations, inhomogeneous spacetimes, modified gravity, and primordial black-hole models.
 • Gravitational entropy and the gravitational arrow of time
 • Schouten–Codazzi gravity
 • Exact and inhomogeneous solutions in General Relativity
 • Primordial black holes and relativistic collapse

Instituto de Ciencias Físicas, Universidad Nacional Autónoma de México (UNAM), Post-Cuernavaca, Morelos, Mexico
 doctoral Researcher
 Postdoctoral research in gravitation and cosmology. 2025

Universidad Iberoamericana, Lecturer in Physics Mexico City, Mexico 2019 – present 7 years
 • Quantum Physics
 • Selected Topics in Astrophysics and Cosmology
 • Vector and Tensor Analysis
 • Electromagnetism

Facultad de Ciencias, Universidad Nacional Autónoma de México (UNAM), Lecturer • Calculus II, III, and IV • Vector Mechanics and Analytical Mechanics • Continuum Mechanics • Thermodynamics • Relativity	Mexico City, Mexico 2016 – 2024 8 years
Facultad de Ciencias, Universidad Nacional Autónoma de México (UNAM), Teaching Assistant • Analytical Mechanics • Calculus I • Advanced Mathematics for Physics	Mexico City, Mexico 2015 – 2016 1 year
Universidad ISEC, Lecturer High-school level physics instruction.	Mexico City, Mexico 2018

Publications

Revisiting the gravitational ‘arrow of time’ Roberto A. Sussman, Sebastián Nájera , Fernando A. Pizaña, Juan Carlos Hidalgo 10.1088/1361-6382/ae6f68 (Classical and Quantum Gravity)	2026
Peculiar velocity fields from analytic solutions of general relativity Roberto A. Sussman, Sebastián Nájera , Fernando A. Pizaña, Juan Carlos Hidalgo 10.1140/epjc/s10052-026-15865-z (The European Physical Journal C)	2026
Gravitational entropy in Petrov Type I spacetimes Maharshi Sarma, Sebastián Nájera , Roberto A. Sussman arxiv.org/abs/2605.20611 (arXiv preprint)	2026
Gravitational entropy of fluids with energy flux Roberto A. Sussman, Sebastián Nájera , Fernando A. Pizaña, Juan Carlos Hidalgo arxiv.org/abs/2509.25007 (arXiv preprint)	2025
Schouten–Codazzi gravity Roberto A. Sussman, Sebastián Nájera 10.1088/1361-6382/ad922e (Classical and Quantum Gravity)	2024

Awards

CONACYT Doctoral Scholarship Scholarship from Consejo Nacional de Ciencia y Tecnología (CONACYT), under the supervision of Roberto A. Sussman.	2020
CONACYT Master’s Scholarship Scholarship from Consejo Nacional de Ciencia y Tecnología (CONACYT), under the supervision of Roberto A. Sussman.	2016–2018

Skills

Research areas: General Relativity, Relativistic Cosmology, Gravitational Entropy, Modified Gravity, Exact Solutions, Inhomogeneous Spacetimes, Primordial Black Holes

Programming and scientific computing: Python, Maple, Mathematica

Technical tools: LaTeX, macOS, Unix/Linux, Microsoft Office

Methods: Exact and semi-analytical methods, Covariant methods, Differential geometry, Relativistic fluid dynamics

Projects

Schouten–Codazzi Gravity

Study of the field equations, integrability conditions, exact solutions, matching problems, perturbations, and compact-object applications of Schouten–Codazzi gravity.

- Exact cosmological and radiating solutions
- Integrability and degree-of-freedom analysis
- Black-hole and no-hair questions

Gravitational Entropy in Inhomogeneous Spacetimes

Investigation of gravitational entropy and the gravitational arrow of time in exact inhomogeneous solutions and relativistic fluids.

- Petrov classification
- Fluids with energy flux
- Exact cosmological models

Exact Models for Primordial Black Holes

Development of exact and semi-analytical General Relativity models that can be compared with numerical primordial black-hole formation calculations.

- LTB and Szekeres models
- Scalar-field models
- Comparison with numerical-relativity approaches

Talks

A non-static inhomogeneous exact solution in $f(T,B)$ gravity

Geometric Foundations of Gravity 2021 (online).

Pancakes as opposed to Swiss Cheese

GRAVITEX 2021 (online).

Szekeres-II models in cosmology

VII Scalar Field Dark Matter Workshop, 2021.

Szekeres-II models in cosmology

XIV DGFM-SMF Workshop, 2021.

Soluciones Exactas en Cosmología - Avances Recientes

Poster presented at LXV Congreso Nacional de Física, 2022.

Neural Networks Classifier for standard and extended cosmologies

Poster presented at CosmoVerse@Lisbon 2023.

Outreach

Relatividad General para principiantes

Public talk at Centro Astronómico Clavius, Universidad Iberoamericana, 2021.

El lado oscuro del Universo

Public talk at Centro Astronómico Clavius, Universidad Iberoamericana, 2020.

Agujeros negros, monstruos cósmicos

Public talk for the Astronomía Básica summer school, Centro Astronómico Clavius, Universidad Iberoamericana, 2020.

El lado oscuro del Universo

Public talk during Astronomy Week at Universidad ITESO, 2020.

Física de los superhéroes

Public talk at Centro Astronómico Clavius, Universidad Iberoamericana, 2020.

Gravitación, Relatividad General y Planilandia

Public talk at Centro Astronómico Clavius, Universidad Iberoamericana, 2019.